# KASHMIR COLLEGE OF ENGINEERING & TECHNOLOGY

Approved by AICTE New Delhi | Affiliated with the University of Kashmir
Campus: Ranbirgarh Narbal, Srinagar - 190017, Jammu & Kashmir | Web: https://kcet.edu.in

## B.Tech in Electronics & Communication - Complete Syllabus

Approved AICTE 4-Year Scheme & Detailed Syllabus Booklet | Category: Syllabus
Curriculum Model: AICTE Model Curriculum / University of Kashmir Ordinances

### Semester 3

* ECE-301: Mathematics-III (Complex Variables & Transforms) (4 Credits)
* ECE-302: Electronic Devices & Circuit Theory (4 Credits)
* ECE-303: Network Analysis & Synthesis (4 Credits)
* ECE-304: Signals & Linear Systems (3 Credits)
* ECE-305: Digital System Design & Verilog (3 Credits)
* ECE-306P: Analog Electronics Circuits Lab (2 Credits)
* ECE-307P: Digital Logic & Simulation Lab (2 Credits)

### Semester 4

* ECE-401: Analog Communication Systems (AM, FM, PM) (4 Credits)
* ECE-402: Linear Integrated Circuits & Op-Amps (4 Credits)
* ECE-403: Electromagnetic Field Theory (EMFT) (4 Credits)
* ECE-404: Microprocessor & Microcontroller Architectures (8085/8086) (3 Credits)
* ECE-405: Probability Theory & Stochastic Processes (3 Credits)
* ECE-406P: Analog Communication Lab (2 Credits)
* ECE-407P: Linear Integrated Circuits Lab (2 Credits)

### Semester 5

* ECE-501: Digital Communication Systems (QPSK, QAM, Coding) (4 Credits)
* ECE-502: Digital Signal Processing (DSP & Filter Design) (4 Credits)
* ECE-503: Control Systems Engineering (3 Credits)
* ECE-504: Embedded Systems & ARM Cortex (3 Credits)
* ECE-PE1: Professional Elective I: Electronic Instrumentation (3 Credits)
* ECE-505P: Digital Communication & DSP Lab (2 Credits)
* ECE-506P: Embedded Systems & Microcontrollers Lab (2 Credits)

### Semester 6

* ECE-601: VLSI Design & CMOS Technology (4 Credits)
* ECE-602: Antenna & Wave Propagation (4 Credits)
* ECE-603: Microwave & Radar Engineering (4 Credits)
* ECE-604: Optical Fiber Communication Systems (3 Credits)
* ECE-PE2: Professional Elective II: Wireless Sensor Networks (3 Credits)
* ECE-605P: VLSI CAD & FPGA Simulation Lab (2 Credits)
* ECE-606P: Microwave & Optical Fiber Lab (2 Credits)

### Semester 7

* ECE-701: Wireless & 5G Cellular Communications (4 Credits)
* ECE-702: Internet of Things (IoT) Architectures & Protocols (3 Credits)
* ECE-PE3: Professional Elective III: Biomedical Electronics (3 Credits)
* OE-701: Open Elective I: Renewable Energy & Power Systems (3 Credits)
* ECE-703P: Capstone Project Stage-I (4 Credits)
* ECE-704I: Industrial Summer Internship Evaluation (2 Credits)

### Semester 8

* ECE-801: Major Capstone Project Stage-II (Hardware / Embedded Defense) (10 Credits)
* ECE-PE4: Professional Elective IV: Satellite & Deep Space Communication (3 Credits)
* OE-801: Open Elective II: Nanoelectronics & Quantum Computing (3 Credits)
* ECE-802V: Comprehensive Technical Viva-Voce (2 Credits)